Delivery Pace Analysis: Freetown, Kenema, and Makeni

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Summary

Objective

To determine a realistic and accurate daily delivery pace to set for each city—Freetown, Kenema, and Makeni—based on past delivery data.

Key Findings

- **Freetown** data shows the highest capacity (64.30 deliveries/day in 2025) with gradual long-term growth averaging 2.21% annually, but with some year-to-year fluctuations.
- **Kenema** data demonstrates significant improvement in average daily deliveries with a 4.49% growth from 2024 to 2025. Kenema's data went from an inconsistent bimodal distribution in 2024 to stable and predictable averaging 40.96 deliveries/day in 2025.
- **Makeni** maintains steady baseline performance at 31.68 deliveries/day with predictable patterns and minimal variability.

Operational Patterns Across All 3 Cities:

- Peak productivity occurs during mid-day hours (12:00-14:00) across all cities.
- Mid-week days demonstrate highest delivery volumes.
- Filtering for delivery accuracy has minimal impact on performance metrics.

Policy Recommendations

2026 Daily Delivery Targets

Targets are set as ranges, with a minimum to sustain productivity and a maximum to reward high achievers without compromising performance quality.

- **Makeni**: 33-50 deliveries per day
- **Kenema**: 43-54 deliveries per day
- **Freetown**: 68-80 deliveries per day

Strategic Considerations

- **Growth Management**: Kenema and Makeni show capacity for continued expansion, while Freetown is slowly reaching sustainable growth management given its mature operational status.
- **Target Setting Framework**: Future targets should incorporate field staff consultation and continuous data monitoring.
- **Fixed Rate Transition**: Consider maintaining fixed targets when average annual growth falls below 2%, operational strain emerges, or distributions stabilize.

Methodology

The datasets used for this analysis were obtained in CSV format. They include: mcc_2025, kcc_2024, kcc_2025, fcc_2021_team_1-3, fcc_2021_team_1-3, fcc_2021_team_4-6, fcc_2022_team_1-3, fcc_2022_team_4-6, fcc_2023_team_1-3, fcc_2023_team_4-6, fcc_2024_internal_staff, fcc_2024_team_1-3,fcc_2024_team_4-6, fcc_2025_rnd1_team_1-3, and fcc_2025_rnd1_team_4-6.

To establish reliable delivery pace benchmarks, the following procedures were applied systematically to each city and year of data:

1. **Comparing Filtered vs. Unfiltered Data:** To understand how accuracy filtering affected the number of daily deliveries enumerators completed.

    - For comparison purposes, two subdatasets were created from the initial dataset—an unfiltered version containing all recorded deliveries and a filtered version restricted to accurate deliveries. Deliveries were classified as accurate if they occurred within 80 meters of the assigned property. This threshold served as a proxy for correct and successfully completed deliveries.
    - The data sets were then grouped by enumerator and date to determine the number of total and accurate deliveries each enumerator completed per day.
    - To ensure results reflected typical working days, extreme values were removed using the Interquartile Range (IQR) method. This prevented unusually high or low delivery counts from skewing the analysis.
    - After cleaning, descriptive statistics were calculated, including the mean, median, standard deviation, interquartile range, and benchmarks at the 80th and 90th percentiles. These provided a balanced picture of typical and high-end performance.
    - Key indicators from the filtered and unfiltered datasets were compared directly. Percentage differences between the two revealed the extent to which accuracy filtering influenced daily delivery averages.

2. **Temporal Analysis:** To understand **when** enumerators were most productive, deliveries were examined across different timeframes:

    - **By time of Day:** Deliveries were categorized into three-hour intervals to pinpoint peak working hours.
    - **By Day of Week:** Average delivery pace was compared across days to detect patterns in weekday versus weekend productivity.
    - **By Day and Time Combined:** Delivery pace was analyzed jointly by day of week and time of day to identify the most productive periods overall.

3. **Visualization of Results:** Key findings were summarized through clear visuals, including time-series plots to show trends, histograms with density curves to illustrate distributions, and bar charts to highlight productivity by day and time brackets.

Note: For the temporal analysis and visualizations, the **unfiltered dataset** was used. This was an executive decision to present delivery pace in its entirety rather than limiting the view to only accurate deliveries. Since errors are an expected part of the process, including them provides a more realistic basis for understanding overall performance and setting daily pace targets.

Replication Across Cities and Years: The full process was applied consistently to each city and each year of available data. This enabled both city-specific insights and cross-city comparisons to identify broader trends and differences.

Contextual Background

The three cities analyzed in this report are at different stages of implementing the property tax reform, which directly affects the scope and structure of available delivery data.

- **Makeni** introduced the property tax reform most recently, in 2024–2025. As a result, only one year of delivery data is available for analysis.
- **Kenema** has two years of delivery data, reflecting its earlier adoption of the reform.
- **Freetown**, with the longest experience, has five years of delivery data, providing the most comprehensive dataset.

Table 1: Sample Sizes: Deliveries and Enumerators by City and Year

City	Year	Total_Deliveries	Total_Enumerators
Makeni	2025	19493	42
Freetown	2021	105998	63
Freetown	2022	104132	63
Freetown	2023	126104	64
Freetown	2024	126100	63
Freetown	2025	96454	63
Kenema	2024	26798	62
Kenema	2025	29839	30

The study comprises 634,918 tax bill deliveries collected across three cities over multiple years. The Freetown sample includes 559,788 observations spanning 2021-2025 (88% of total sample), with an average of 63-64 enumerators anually . The Kenema sample contains 56,637 observations across 2024-2025, with number of deliveries deliveries increasing 11.4% (from 26,798 to 29,839) while the total number of enumerators decreased from 62 to 30 individuals. The Makeni sample consists of 19,493 deliveries from 2025 with a total number of 42 enumerators, the smallest dataset. The variation in sample sizes and total number of enumerators across cities reflects differences in population density, administrative capacity, and delivery methodologies that should be considered during cross-city comparisons.

Note: The reported sample sizes—especially the total number of deliveries—may underestimate the actual number of deliveries completed. Datasets may be missing data due to technological issues and/or instances where enumerators did not submit their deliveries via delivery apps.

In addition to differences in data length and sample size, the underlying data structures vary by city due to the use of different versions of the MOPTAX system:

- **Makeni's** data was captured entirely in **MOPTAX 2.0**.
- **Kenema's** data spans both **MOPTAX 1.0 and 2.0**, creating a hybrid structure.
- **Freetown's** data was recorded exclusively in **MOPTAX 1.0**.

These variations matter because the two MOPTAX versions create datasets that differ in column names, value formats, and certain metrics. Consequently, the analysis required careful adjustments to account for system-specific differences and ensure comparability across cities.

Moreover, Freetown's 2025 dataset contains additional complexities, as it includes two rounds of delivery. This analysis focuses only on Round 1, which primarily consisted of residential (RDN) deliveries and excluded business licenses. Business license deliveries often take a different amount of time than residential deliveries, thus this data irregularity could distort delivery pace outcomes. As such, results from Freetown 2025 should be interpreted with this context in mind.

Finally, also important to note is that most typical delivery period work days span from Monday or Tuesday through Friday or Saturday, and should almost always fall within standard operational hours of 8:00 a.m. to 5:30 p.m. This has implications for deriving insights from the temporal analysis conducted.

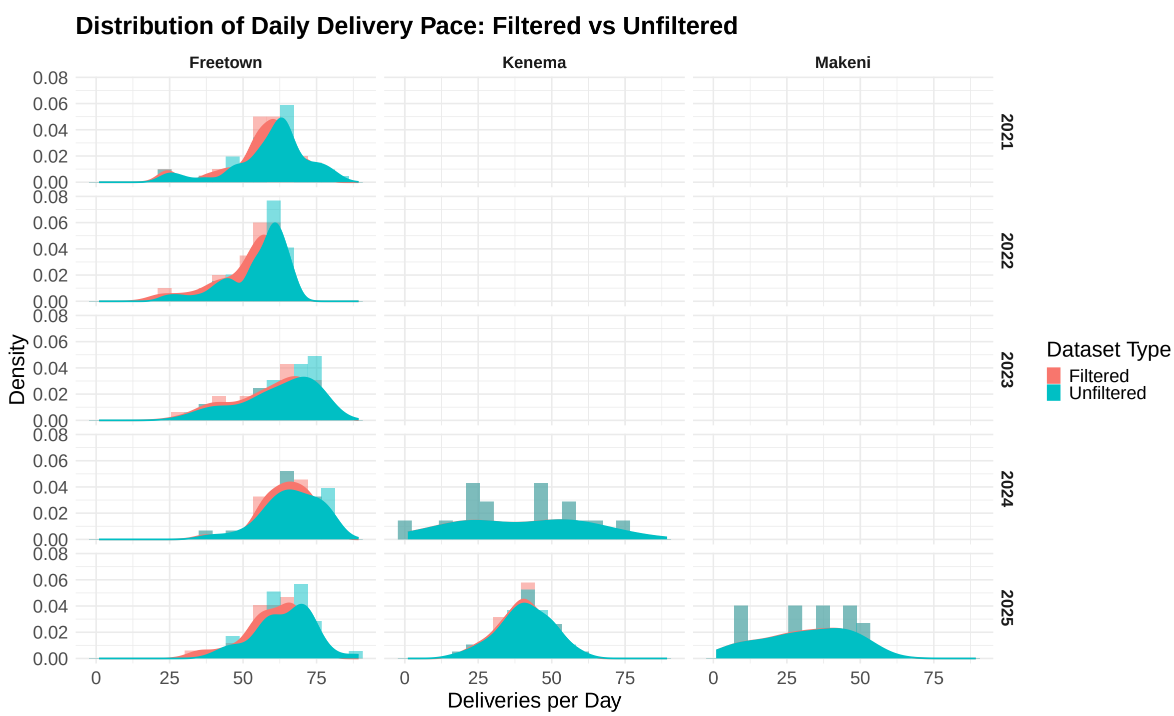
Results & Discussion

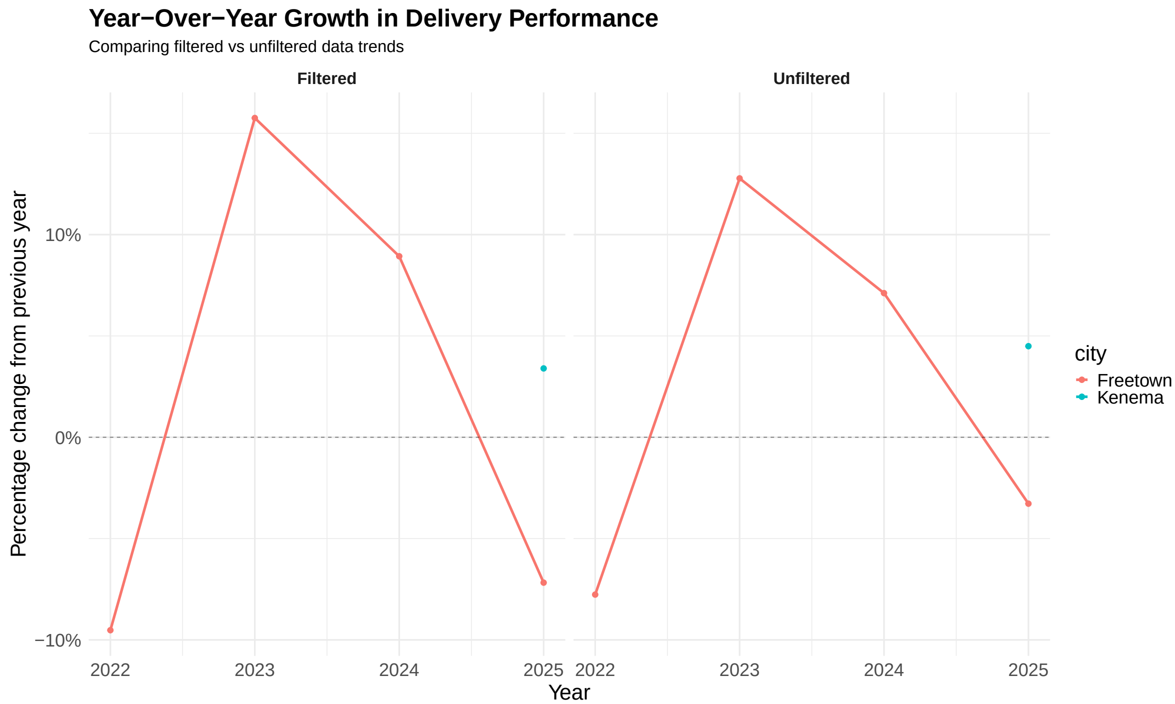
Filtered vs. Unfiltered Data Comparison

This section evaluates how filtering for accurate deliveries affects the average delivery daily pace and highlights trends over multiple years for Makeni, Kenema, and Freetown.

Table 2: Summary of Daily Deliveries by City, Year, and Dataset

City	Year	Dataset	Mean	SD	Median	P80	P90
Freetown	2021	Filtered	56.41	12.28	59.13	64.24	69.87
Freetown	2021	Unfiltered	59.67	12.99	62.56	67.88	74.54
Freetown	2022	Filtered	51.04	10.65	55.42	59.42	61.04
Freetown	2022	Unfiltered	55.04	10.29	59.43	62.66	65.50
Freetown	2023	Filtered	59.08	12.46	61.31	71.05	72.56
Freetown	2023	Unfiltered	62.07	12.78	65.29	73.65	75.24
Freetown	2024	Filtered	64.35	8.59	65.02	71.88	73.77
Freetown	2024	Unfiltered	66.48	9.74	66.37	75.57	78.38
Freetown	2025	Filtered	59.74	9.66	61.37	67.41	69.68
Freetown	2025	Unfiltered	64.30	10.05	66.66	71.74	74.15
Kenema	2024	Filtered	38.83	21.50	45.69	56.71	62.98
Kenema	2024	Unfiltered	39.20	21.65	46.07	57.21	63.47
Kenema	2025	Filtered	40.15	8.77	40.43	48.38	50.19
Kenema	2025	Unfiltered	40.96	9.17	41.10	48.50	51.55
Makeni	2025	Filtered	31.28	14.59	32.60	45.12	48.10
Makeni	2025	Unfiltered	31.68	14.85	33.09	46.00	48.63





Makeni (2025)

In Makeni, the analysis shows that the mean daily deliveries were 31.68 in the unfiltered dataset compared to 31.28 in the filtered dataset, representing a minimal 1.24% difference. The median values were 33.09 and 32.60 deliveries, respectively, while the standard deviation ranged from 14.6 to 14.8 deliveries, indicating moderate but acceptable variability. High-performance benchmarks, as reflected in the 80th and 90th percentiles, reached 45–46 and 48+ deliveries per day, respectively. These findings indicate that filtering has minimal impact on overall performance, suggesting that errors or anomalies in the dataset do not meaningfully distort operational trends.

The distribution analysis confirms a normal distribution with a peak around 30–35 deliveries, with filtered and unfiltered datasets closely overlapping. The narrower delivery range (approximately 15–50 deliveries) with concentration around 25–40 deliveries supports stable but moderate-capacity operations. Overall, Makeni demonstrates stable average daily deliveries, with consistent average output and predictable peaks.

Kenema (2024-2025)

In Kenema, the 2024–2025 comparison shows that the mean daily deliveries increased slightly from 38.83 in the filtered dataset in 2024 to 40.15 in 2025, representing a 3.40% increase, while the unfiltered dataset shows a comparable 4.49% increase over the same period. The median daily deliveries remained similar, though the standard deviation decreased sharply from approximately 21.5 in 2024 to 9.0 in 2025, reflecting a marked improvement in consistency and reduced day-to-day variability. The 80th and 90th percentiles indicate that high-output days became more predictable, even as extreme peaks were slightly lower. These trends suggest that Kenema's operations in 2025 were smoother and more reliable compared to 2024. Overall, filtering has a minimal effect on the assessment of operational performance but ensures that the analysis reflects accurate, error-free data.

The distribution analysis shows that Kenema exhibits a substantial transformation between its 2024 and 2025 distributions. In 2024, the distribution was bimodal, with peaks around 20 and 45 deliveries and a broad performance range spanning 0–80 deliveries. This pattern reflected inconsistency, with significant low-performance clustering around 0–20 deliveries. By 2025, operations stabilized significantly. The distribution transformed into a concentrated single-peaked normal pattern centered around 40 deliveries, with a narrower range of 15–60 deliveries. Filtering impact remained low, with filtered and unfiltered distributions closely aligned. The transformation indicates that Kenema's operations became more consistent, eliminating extreme low-performance days and supporting predictable scheduling overtime.

Freetown (2021-2025)

In Freetown, analysis across multiple years demonstrates both growth and fluctuations in operational performance. The mean daily deliveries in filtered dataset were 56.41 in 2021, 51.04 in 2022, 59.08 in 2023, 64.35 in 2024, and 59.74 in 2025. In unfiltered dataset, the mean deliveries were slightly higher each year, ranging from 59.67 in 2021 to 64.30 in 2025. The median and high-performance thresholds followed similar trends, reflecting growing peak capacity over time.

The year-over-year performance changes provide additional insight. In the filtered dataset, Freetown experienced a decrease of 9.52% from 2021 to 2022, followed by strong growth of 15.75% in 2022–2023 and 8.93% in 2023–2024, before declining 7.17% in 2024–2025. Unfiltered data shows a similar patter with slightly smaller fluctuations, with increases of 12.77% from 2022–2023 and 7.11% from 2023–2024, followed by a decrease of 3.28% in 2024–2025. The average annual change rates indicate gradual long-term growth, with 1.997% for filtered data and 2.212% for unfiltered data.

These results indicate that, while daily averages fluctuate, Freetown has achieved a general upward trend in operational capacity over the years. Filtering reduces the impact of anomalies on the data but does not materially change the observed trends, suggesting that the improvements and declines reflect true operational performance rather than delivery errors.

Freetown demonstrates consistently high delivery volumes and generally normal distribution shapes across all years. The distributions were predominantly single-peaked and normal, with minor right skew in 2021. Filtered and unfiltered dataset closely overlapped in all years, though small differences in tail regions. Year-over-year, Freetown shows clear distribution evolution. In 2021–2022, the distribution shifted slightly left, reflecting a modest decrease in performance. From 2022–2025, distributions shifted gradually rightward, demonstrating improvements in daily delivery performance, with peak activity consistently concentrated in 45–75 deliveries. The mode consistently aligns with the mean, confirming symmetric, normal distributions, and standard deviations remained moderate, supporting predictable delivery patterns.

Cross-City Insights

Across cities, several patterns emerge. Freetown demonstrates gradual long-term growth, with increasing mean deliveries and higher high-performance thresholds over time, though fluctuations in certain years indicate periods of operational adjustment or demand variation. Kenema shows measurable improvements in 2025 compared to 2024, with higher average deliveries and significantly reduced variability, indicating more consistent and reliable daily operations. Makeni, while only available for 2025, shows stable performance with predictable peaks and minimal difference between filtered and unfiltered data.

Overall, filtered data confirms that inaccurate deliveries do not significantly distort average daily delivery pace, while unfiltered data reflects the full operational reality. Across all cities, the differences between filtered and unfiltered datasets are small, reinforcing that the analysis of mean, median, and peak performance provides a reliable picture of delivery operations. Year-over-year and average annual change rates highlight the evolution of capacity, with Kenema and Freetown showing positive growth trends and improved operational stability.

Time of Day & Day of Week Analysis

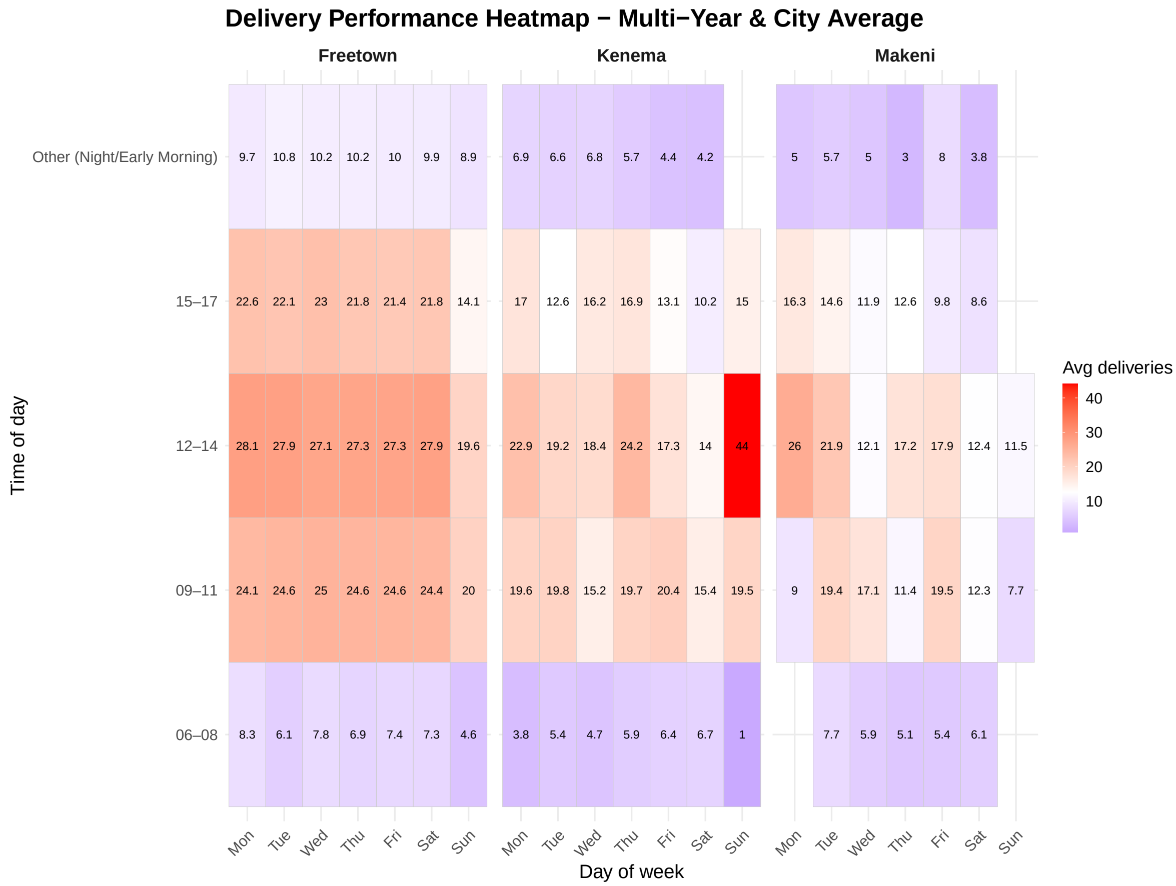
This section analyzes average delivery pace by **day of week** and **time of day**, highlighting cross-year and cross-city trends.

Table 3: Average Delivery Pace by Time of Day Across All Cities and Years

City	Time_of_Day	Average	Years	Min	Max
Freetown	06–08	7.16	5	6.49	8.67
Freetown	09–11	24.45	5	20.29	27.29
Freetown	12–14	27.53	5	24.10	30.88
Freetown	15–17	22.05	5	19.08	23.85
Freetown	Other (Night/Early Morning)	10.10	5	7.91	13.94
Kenema	06–08	5.42	2	4.87	5.97
Kenema	09–11	19.44	2	18.00	20.87
Kenema	12–14	20.90	2	20.55	21.24
Kenema	15–17	14.87	2	14.71	15.02
Kenema	Other (Night/Early Morning)	5.72	2	4.72	6.72
Makeni	06–08	5.91	1	5.91	5.91
Makeni	09–11	14.94	1	14.94	14.94
Makeni	12–14	18.39	1	18.39	18.39
Makeni	15–17	12.89	1	12.89	12.89
Makeni	Other (Night/Early Morning)	4.97	1	4.97	4.97

Table 4: Average Delivery Pace by Day of Week Across All Cities and Years

City	Day_of_Week	Average	Years	Min	Max
Freetown	Sun	39.78	5	23.25	53.36
Freetown	Mon	65.23	5	58.42	73.73
Freetown	Tue	64.55	5	59.47	71.77
Freetown	Wed	65.26	5	59.05	70.60
Freetown	Thu	63.85	5	59.08	67.73
Freetown	Fri	64.44	5	57.88	68.32
Freetown	Sat	64.11	5	57.50	69.69
Kenema	Sun	49.50	1	49.50	49.50
Kenema	Mon	50.11	2	38.80	61.43
Kenema	Tue	40.37	2	37.73	43.01
Kenema	Wed	33.69	2	21.03	46.36
Kenema	Thu	53.74	2	45.70	61.79
Kenema	Fri	40.00	2	36.34	43.66
Kenema	Sat	27.63	2	21.74	33.52
Makeni	Sun	13.80	1	13.80	13.80
Makeni	Mon	43.57	1	43.57	43.57
Makeni	Tue	48.66	1	48.66	48.66
Makeni	Wed	29.22	1	29.22	29.22
Makeni	Thu	27.44	1	27.44	27.44
Makeni	Fri	36.30	1	36.30	36.30
Makeni	Sat	23.41	1	23.41	23.41



Makeni

2025 Performance

In Makeni's single year of observation (2025), the highest-performing day is Tuesday, averaging 48.66 deliveries per enumerator, followed by Monday at 43.57 and Friday at 36.30. Wednesday and Thursday are moderate-performance days with averages of 29.22 and 27.44 deliveries respectively, while Saturday delivers 23.41 on average. Sunday represents the lowest activity at 13.80 deliveries, likely reflecting reduced operational hours or staffing on this traditional rest day.

Examining the time-of-day performance in 2025, the 12:00–14:00 period emerges as the most productive window at 18.39 deliveries per enumerator, followed by 09:00–11:00 with 14.94 deliveries, and 15:00–17:00 with 12.89 deliveries. Early morning (06:00–08:00) and night/early morning periods remain predictably low at 5.91 and 4.97 deliveries respectively.

Average Performance

Since only one year of data is available, the 2025 figures represent Makeni's averages. The combination of day-of-week and time-of-day analysis indicates that Makeni's peak operational capacity is concentrated in early-to-mid-week and midday periods, particularly Tuesday midday hours. The significant drop-off from Tuesday's peak to other weekdays (up to 73% lower on Sunday) suggests potential opportunities for more consistent resource allocation and operational standardization. Without multi-year data, it remains unclear whether these patterns represent stable operational norms or a single-year snapshot that may not reflect longer-term trends.

Kenema

2024 Performance

In 2024, Kenema's operations showed a distinctive pattern with Thursday and Monday as the highest-performing days. Thursday averaged 61.79 deliveries per enumerator, and Monday averaged 61.43 deliveries. Tuesday performed moderately at 43.01 deliveries, while Friday averaged 43.66 deliveries. Wednesday and Saturday were notably weak, averaging just 21.03 and 21.74 deliveries respectively—representing a 65-66% decline from the peak days. There are no recorded deliveries on Sundays for Kenema's 2024 data.

Time-of-day patterns in 2024 showed the 12:00–14:00 period leading at 21.24 deliveries per enumerator, followed by 09:00–11:00 at 20.87 deliveries and 15:00–17:00 at 15.02 deliveries. Early morning (06:00–08:00) averaged 5.97 deliveries, while night/early morning periods averaged 6.72 deliveries.

2025 Performance

In 2025, Kenema's operational patterns shifted significantly. Sunday emerged as an important day with 49.50 deliveries per enumerator. Wednesday rebounded dramatically to 46.36 deliveries—more than doubling from its 2024 low. Thursday remained relatively strong at 45.70 deliveries, though down from 2024's peak. Monday declined to 38.80 deliveries (a 37% drop from 2024), Tuesday fell to 37.73 deliveries, and Friday decreased to 36.34 deliveries. Saturday remained the lowest weekday at 33.52 deliveries but improved 54% from 2024.

Time-of-day performance in 2025 showed some compression. The 12:00–14:00 period averaged 20.55 deliveries per enumerator (slightly lower than 2024), 09:00–11:00 dropped to 18.00 deliveries (14% decline), and 15:00–17:00 remained stable at 14.71 deliveries. Early morning (06:00–08:00) decreased to 4.87 deliveries, while night/early morning periods fell to 4.73 deliveries.

Average Performance (2024-2025)

Across both years, Kenema averages reveal moderate overall performance with high variability. Monday averages 50.11 deliveries per enumerator across the two years (though this masks the 37% year-over-year decline). Thursday averages 53.74 deliveries—the highest across both years. Tuesday averages 40.37 deliveries, Wednesday averages 33.69 deliveries (despite the 121% improvement from 2024 to 2025), Friday averages 40.00 deliveries, and Saturday averages 27.63 deliveries. Sunday's single-year figure of 49.50 cannot be averaged meaningfully.

For time-of-day averages, the 12:00–14:00 window averages 20.90 deliveries per enumerator across both years, 09:00–11:00 averages 19.44 deliveries, 15:00–17:00 averages 14.87 deliveries, 06:00–08:00 averages 5.42 deliveries, and night/early morning periods average 5.72 deliveries.

The dramatic year-over-year variations in day-of-week performance (up to 121% increases or 37% decreases) illlustrates a significant operational restructuring from 2024-2025 delivery in Kenema. The relative stability in time-of-day patterns, by contrast, indicates more consistent intra-day scheduling practices even as weekly allocation changes.

Freetown

2021 Performance

In 2021, Freetown's weekday performance showed Friday leading at 64.71 deliveries per enumerator, followed by Wednesday at 62.98, Thursday at 61.24, Monday at 61.62, Tuesday at 59.47, and Saturday at 59.98. Sunday recorded 37.50 deliveries.

Time-of-day performance in 2021 showed the 12:00–14:00 period at 24.10 deliveries per enumerator, 09:00–11:00 at 20.29 deliveries, 15:00–17:00 at 19.08 deliveries, 06:00–08:00 at 6.49 deliveries, and night/early morning at 7.91 deliveries.

2022 Performance

In 2022, weekday patterns shifted slightly with Monday rising to 63.72 deliveries per enumerator, Tuesday at 64.77, Wednesday at 64.04, Thursday at 59.08, Friday at 57.88, and Saturday at 57.50. Sunday improved notably to 44.35 deliveries.

Time-of-day performance in 2022 showed the 12:00–14:00 period at 26.26 deliveries, 09:00–11:00 at 23.43 deliveries, 15:00–17:00 at 20.16 deliveries, 06:00–08:00 at 6.70 deliveries, and night/early morning at 8.62 deliveries.

2023 Performance

In 2023, Monday reached 64.62 deliveries per enumerator, Tuesday peaked at 71.77, Wednesday at 70.60, Thursday at 67.73, Friday at 68.32, and Saturday at 69.69. Sunday increased to 53.36 deliveries—the highest Sunday performance across all years.

Time-of-day performance in 2023 showed the 12:00–14:00 period at 28.20 deliveries, 09:00–11:00 at 25.04 deliveries, 15:00–17:00 at 23.85 deliveries, 06:00–08:00 at 8.67 deliveries, and night/early morning at 13.94 deliveries.

2024 Performance

In 2024, Monday delivered 58.42 deliveries per enumerator, Tuesday 60.35, Wednesday 59.05, Thursday 60.74, Friday 61.78, and Saturday 64.65. Sunday dropped sharply to 23.25 deliveries—the lowest Sunday performance across all years.

Time-of-day performance in 2024 showed the 12:00–14:00 period at 27.02 deliveries, 09:00–11:00 at 24.57 deliveries, 15:00–17:00 at 21.04 deliveries, 06:00–08:00 at 6.95 deliveries, and night/early morning at 10.11 deliveries.

2025 Performance

In 2025, Monday surged to 73.73 deliveries per enumerator—the highest single day-of-week performance across all years and cities. Tuesday reached 66.42, Wednesday 70.65, Thursday 70.49, Friday 68.49, and Saturday 69.08. Sunday data is not provided for 2025.

Time-of-day performance in 2025 showed the 12:00–14:00 period at 30.88 deliveries—the highest midday performance across all years. The 09:00–11:00 period reached 27.29 deliveries, 15:00–17:00 at 21.09 deliveries, 06:00–08:00 at 7.12 deliveries, and night/early morning at 10.02 deliveries.

Average Performance (2021-2025)

Across five years, Freetown's averages demonstrate consistently high productivity with moderate variation. Monday averages 65.23 deliveries per enumerator (ranging from 58.42 to 73.73—a 26% spread), Tuesday averages 64.55 deliveries (59.47 to 71.77—a 21% spread), Wednesday averages 65.26 deliveries (59.05 to 70.65—a 20% spread), Thursday averages 63.85 deliveries (59.08 to 67.73—a 15% spread), Friday averages 64.44 deliveries (57.88 to 68.32—an 18% spread), and Saturday averages 64.11 deliveries (57.50 to 69.69—a 21% spread). Sunday averages 39.78 deliveries across the four years with data (23.25 to 53.36—a dramatic 129% spread indicating evolving weekend operation policies).

Time-of-day averages across the five years show the 12:00–14:00 period at 27.53 deliveries per enumerator (24.10 to 30.88—a 28% increase trajectory), 09:00–11:00 at 24.45 deliveries (20.29 to 27.29—a 35% increase trajectory), 15:00–17:00 at 22.05 deliveries (19.08 to 23.85—a

25% increase trajectory), 06:00–08:00 at 7.16 deliveries (6.49 to 8.67—a 34% spread), and night/early morning at 10.10 deliveries (7.91 to 13.94—a 76% spread).

The year-over-year trajectory shows Freetown's operations strengthening over time, with 2025 representing peak performance in most metrics. The relative stability in weekday averages (Monday through Saturday showing less than 3% variation from each other) demonstrates exceptional operational consistency. The gradual improvements in midday productivity (28% increase from 2021 to 2025) suggest successful operational optimization and capacity building over the five-year period.

Cross-City Insights

When comparing cities, several consistent patterns emerge. Firstly, mid-day hours (12:00–14:00) consistently represent the most productive window across all cities and years. Morning (09:00–11:00) and afternoon (15:00–17:00) periods complement these peaks, enabling near-continuous operational productivity. Whiile early morning (06:00–08:00) and night shifts consistently show low activity, which is to be expected given operational policies restricting work during these hours.

Mid-week or early-week days generally deliver the highest volume, while Sundays consistently record the lowest deliveries.

Conclusions & Policy Recommendations

The analysis reveals distinct delivery pace profiles across the three cities, with each demonstrating unique strengths and improvement opportunities. Freetown demonstrates the highest average daily deliveries with an average of 59.74 daily deliveries, though it experiences some performance fluctuations that require careful management. Kenema shows the most encouraging improvement trajectory, increasing from 38.83 deliveries in 2024 to 40.15 deliveries in 2025 while significantly reducing day-to-day variability. Makeni maintains stable and predictable performance with 31.28 daily deliveries, providing a solid foundation for sustainable growth. These findings inform targeted recommendations designed to set sustainable daily delivery paces across all three cities in the 2025-2026 fiscal year and beyond.

Proposed City-Specific Delivery Pace Targets for the 2026 Fiscal Year

Table 5: Recommended 2026 Delivery Pace Targets Setting by City

City	Year	Baseline_Mean	Projected_Mean	SD	Days	CI_Low	CI_High	High_Perf
Makeni	2026	31.68	33.10	14.85	16	24.83	41.37	50.81
Kenema	2026	40.96	42.80	9.17	41	39.78	45.83	53.86
Freetown	2026	66.48	67.95	9.74	33	64.42	71.48	80.11

Proposed City-Specific Delivery Pace Targets for the 2026 Fiscal Year

Targets are established as ranges, providing both lower and upper benchmarks to guide enumerator performance. The minimum threshold functions as a safeguard to ensure consistent productivity across the team, discouraging underperformance and reinforcing accountability. It also serves as a critical reference point for the design of delivery zones, ensuring workloads are distributed fairly and remain manageable for all enumerators. Conversely, the maximum threshold is intended to recognize and reward the contributions of higher-achieving enumerators while at the same time preventing risks associated with excessive workloads, such as fatigue, mistakes, or declining data quality. This approach strikes a balance between motivating steady improvement, accommodating varying capacities, and maintaining high performance standards without compromising quality.

Makeni

The 2025 unfiltered data for Makeni indicates an average daily delivery pace of **31.68 deliveries per day**, establishing the foundation for 2026 target calculations. Given the absence of multi-year historical data for Makeni, Kenema's average annual delivery pace growth rate of **4.49%** was used as a proxy metric.

- **Projected 2026 baseline target:** 33.10 deliveries per day
- **95% Confidence Interval (CI):** 24.83 – 41.37 deliveries
- **High-performance threshold (90th percentile):** 50.16 deliveries

Recommended 2026 delivery range: 33 – 50 deliveries per day.

Kenema

Kenema's 2025 unfiltered data has a mean of **40.96 deliveries per day**. Projecting forward with the established growth trajectory of **4.49%**, Kenema's 2026 baseline target reaches **42.80 deliveries per day**.

- **95% CI for baseline:** 39.78 – 45.83 deliveries

- **High-performance threshold (90th percentile):** 53.86 deliveries.

Recommended 2026 delivery range: 43 – 54 deliveries per day

Freetown

Freetown's delivery operations represent the most established and highest-capacity center among the three locations, with 2024 baseline performance averaging **66.48 deliveries per day**. The city's multi-year performance analysis yields a conservative annual growth rate of **2.21%**, reflecting the mature nature of operations and the challenges associated with scaling already high-volume activities.

- **Projected 2026 baseline target:** 67.95 deliveries per day
- **95% CI:** 64.42 – 71.48 deliveries
- **High-performance threshold (90th percentile):** 80.11 deliveries.

Recommended 2026 delivery range: 68 – 80 deliveries per day

Note: 2024 is used as a baseline for Freetown given the anomalies in the 2025 data.

Policy Considerations For Future Delivery Pace Target Setting

Establishing delivery pace targets is not an exact science. While data provides essential benchmarks, effective target-setting must incorporate broader practical considerations, including operational realities and staff input.

- **Field consultation:** Feedback from field staff is critical to ensure that targets remain realistic, achievable, and reflective of actual working conditions.
- **Ongoing monitoring:** Targets should be informed by continuous tracking of annual changes in average daily delivery pace, with attention to percentage growth patterns. This ensures projections are aligned with both potential and capacity.

Importantly, there will come a point where adopting a **fixed delivery pace target**—rather than pursuing continual increases—becomes the most sustainable and effective approach. This stage has not yet been reached in Kenema and Makeni, where post–property tax reform implementation still presents opportunities for growth. However, in more mature systems such as Freetown's, conditions for fixed targets may emerge sooner. This is important because while incremental increases can encourage progress, targets that exceed realistic capacity risk undermining morale if perceived as unattainable. Higher unrealistic delivery pace targets and compounded growth rates—even as modest as 4–5% annually—can surpass actual infrastructure or staffing improvements prompting errors, rework, and inefficiencies, and negating intended productivity gains.

Indicators for Maintaining a Fixed Rate

- If year-to-year growth levels off and significant gains are no longer observed, maintaining a fixed baseline may be more meaningful than ongoing adjustments.
- If filtered and unfiltered data show minimal divergence, indicating fewer outliers and a stable system, a fixed target can help consolidate consistent performance.
- If year-over-year increases in average daily delivery pace fall below 2%.
- If delivery pace distributions stabilize, with little upward movement in median or peak values.
- If signs of operational strain emerge (e.g., rising error rates, absenteeism, or delays).
- If staff consistently meet but do not exceed targets despite additional training or support.